

INSTITUTE FOR COMPUTER SCIENCE, GÖTTINGEN

Privacy-aware Environment Map

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Motivation & Introduction

Motivation: Privacy in Robotics

- Robots increasingly operate in human-occupied spaces such as labs, offices, and homes, where they continuously sense their surroundings.
- It creates a serious privacy threat because these systems can perceive, record, and analyze the environments in which people live and work.
- If such information is used improperly, the consequences could include exposure of personal information, and misuse of sensitive data.
- This motivates the urgent need for privacy-aware robotic systems.

Privacy is a Big Concern



Cleaning Effectiveness and Data Privacy Issues of Automatic Vacuum Robots



Data Privacy concerns erupt with Robotic Vacuums



Goal: Develop a system that constructs an environment map of the robot and highlights privacy-sensitive objects within it.

Image Source: Matic Robots

Care-O-Bot

- Care-O-Bot is a family of service robots developed by Fraunhofer IPA.
- It assists individuals in performing a wide range of routine daily activities.
- It can move autonomously, sense its surroundings, and communicate with people.
- Different generations of Care-O-Bot have been developed with improved mobility, sensing, and manipulation capabilities.



Image Source: Fraunhofer IPA

Privacy Issues with Care-O-Bot 4

- Care-O-Bot 4 may collect sensitive personal data through cameras, microphones and other sensors.
- It may observe and process information about users, their activities, and interactions.
- It may monitor people and their activities in private spaces.
- Information collected by the robot may need to be stored, creating risks if it is not properly protected.
- Poor security could allow unauthorized people to access the robot or the data it collects.



Image Source: Fraunhofer IPA

Related Work

SLAM for Mobile Robots

- 2D approaches (e.g., grid-based methods like GMapping, or graph-based methods like Cartographer (Hess et al., 2016)) build occupancy-grid maps from planar LiDAR scans.
- 3D approaches (e.g., ORB-SLAM (Mur-Artal et al., 2015), OctoMap (Hornung et al., 2013)) reconstruct volumetric or feature-based maps from cameras and depth sensors.
- Semantic SLAM extends either approach by associating spatial locations with recognized objects.

Object Detection for Robotic Perception

- YOLO provides real-time single-stage object detection suitable for robotic perception.
- Faster R-CNN represents a two-stage detection approach.
- YOLO has been widely integrated into robotics for object recognition and scene understanding.

Semantic & Privacy-Aware Mapping

- Semantic maps associate detected objects/classes with locations in the environment.
- Privacy-aware robotics addresses sensitive information encountered by robots in human-occupied spaces.
- Existing work motivates robots to recognize and appropriately handle privacy-sensitive information.

Positioning of Our Work

- GMapping → geometric environment map
- YOLOv8n → privacy-sensitive object detection
- RGB-D + camera intrinsics + TF → map-frame object localization
- RViz markers → unified privacy-aware environment map

Our contribution: combining classical SLAM, deep-learning detection, depth-based localization, and privacy-sensitive semantic markers in one map.

Technology Stack

Robot / Simulation

- Ubuntu
- Robot Operating System (ROS)
- Gazebo

Visualization / Perception

- Python
- YOLO
- RViz



GAZEBO

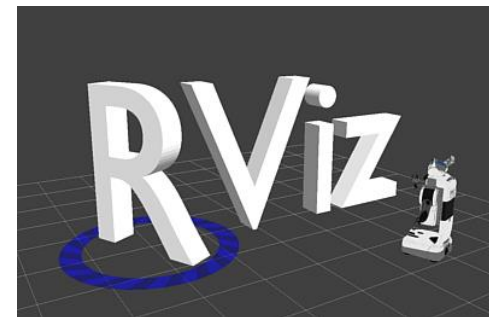
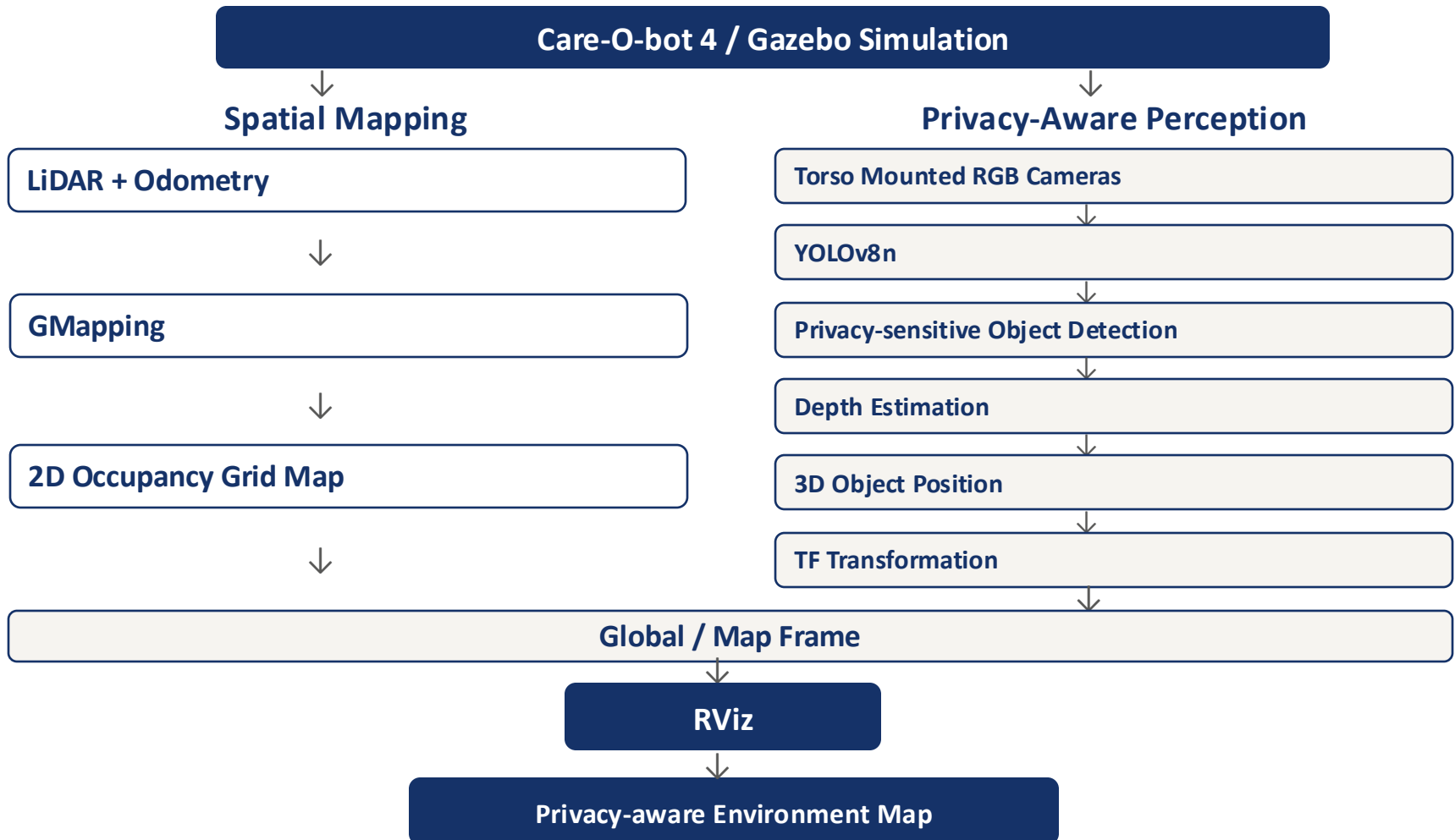


Image Source: Respective Software Website

Overall System Flow



Robot Sensors Used

PLANAR LiDAR

- 3 laser scanners: Front / Left / Right
- Raw topics:
/base_laser_front/scan
/base_laser_left/scan
/base_laser_right/scan
- Combined into: /scan_unified
- **Purpose:** Environment mapping with GMapping SLAM

LEFT + RIGHT RGB CAMERAS

- Torso-mounted RGB cameras
- Left:
/torso_cam3d_left/rgb/image_raw
- Right:
/torso_cam3d_right/rgb/image_raw
- **Purpose:** YOLO-based detection of privacy-sensitive objects
- Two detector instances run in parallel

RGB-D DEPTH CAMERAS

- Registered depth for left/right RGB cameras
- Left:
/torso_cam3d_left/depth_registered
- Right:
/torso_cam3d_right/depth_registered
- **Purpose:** Metric depth for detected objects and 3D localization

SUPPORTING DATA (not physical sensors)

CameraInfo → camera intrinsics (fx, fy, cx, cy)

TF + robot pose → transform camera-frame points into /map

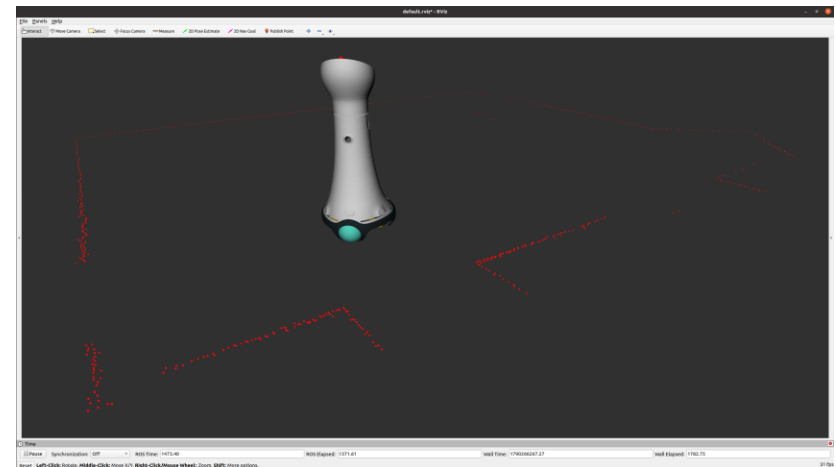
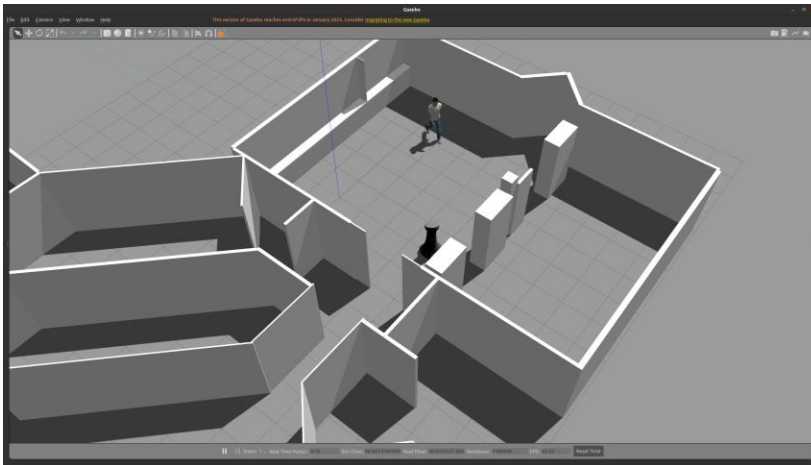
Odometry (/odom_combined) → used by SLAM/TF configuration

LiDAR → GMapping → 2D Map

RGB + RGB-D → YOLO + Depth → 3D Object Position

LiDAR Setup

- Care-O-Bot 4 provides three laser scanners: Front LiDAR, Left LiDAR, Right LiDAR.
- The three LiDAR sensors provide planar distance measurements of surrounding objects and obstacles.
- They continuously provide range measurements as the robot moves through the environment.
- Their measurements are combined into the unified ROS topic: /scan_unified.
- The unified laser scan is the input to the Simultaneous Localization and Mapping (SLAM) system.



Simultaneous Localization and Mapping (SLAM)

- SLAM enables a robot to locate itself while simultaneously mapping its surroundings.
- The robot uses sensor data to discover walls, obstacles, and other structures.
- The robot creates a map and locates itself on that map at the same time.
- Simple analogy for a robot: “Where am I, and what does the world around me look like?”
- SLAM is widely used in robotics, autonomous vehicles, drones, AR/VR, and robot vacuum cleaners.



Image Source: MathWorks

GMapping

- GMapping is a SLAM algorithm based on Rao-Blackwellized Particle Filter (RBPF).
- It uses **laser scanner/LiDAR** to measure distances to walls and obstacles.
- It uses **odometry** to estimate how the robot has moved.
- GMapping combines these measurements (LiDAR data + Odometry) to build a **2D occupancy grid map**.
- At the same time, it estimates the **robot's position (pose)** in the environment.

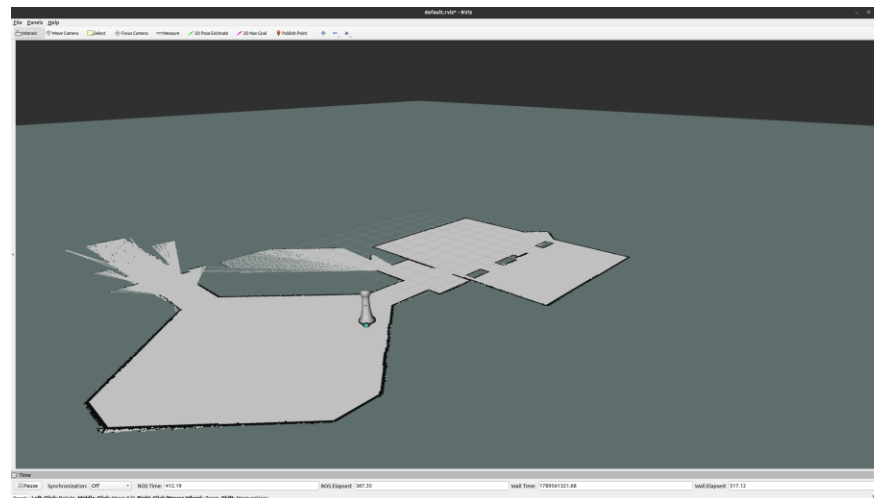
LiDAR Data + Odometry + Robot Pose → GMapping → 2D Occupancy Map



Building Environment Map using GMapping (SLAM)

- 1) Robot starts in the simulated apartment.
- 2) LiDAR continuously measures surrounding structures.
- 3) Robot is moved through the environment using keyboard teleoperation.
- 4) GMapping combines motion information and laser observations.
- 5) The estimated trajectory is used to place observations consistently in the map.
- 6) The **occupancy grid** grows as **new areas** are observed.

Start Simulation → Drive Robot → LiDAR Observations + Odometry → GMapping → Growing Occupancy Grid



Privacy-Sensitive Object Classes



Human Face

Can directly identify a person, making it personally identifiable information.



Display Monitor

May show emails, messages, personal accounts, or other confidential information.



ID Card

May expose a name, photograph, identification number, or other identifying credentials.



Documents

May contain personal, confidential, academic, or financial information.



Medicines

Packaging may reveal a person's health conditions or treatments.



Laptop Screen

May display private communications, personal data, or confidential information.



Car Number Plate

Contains a unique registration identifier that may trace a vehicle and its owner.



Debit / Credit Card

May expose financial information and identifiers tied to a person's account.

YOLO Model

Model

YOLOv8n (Nano)

Data Source

Roboflow

Classes

8 privacy-sensitive classes

Dataset

3,256 training + 467 validation images

Task & Annotation Format

Object Detection - Bounding Box

Training Approach

Transfer Learning

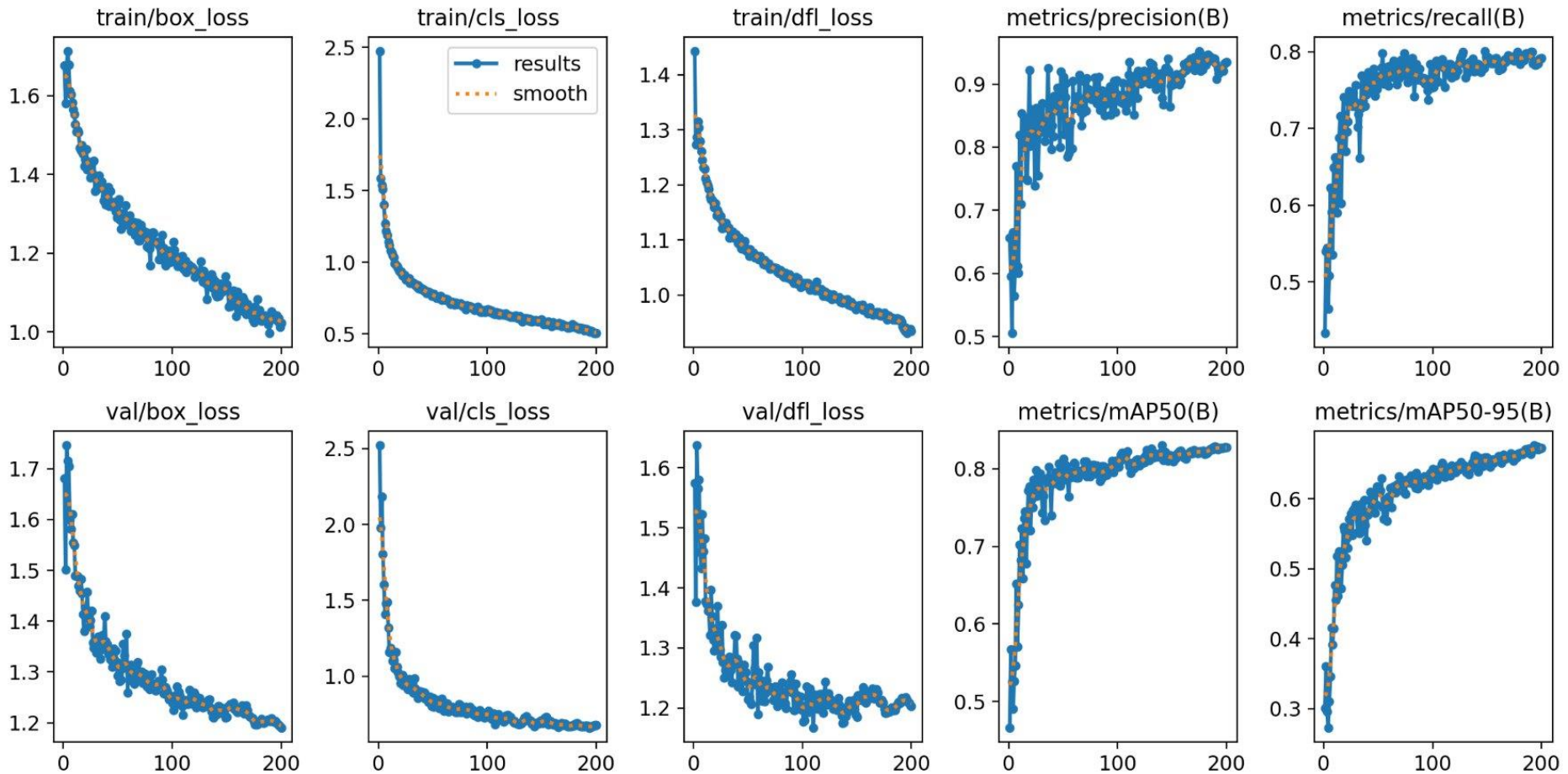
Training Epochs

200 epochs

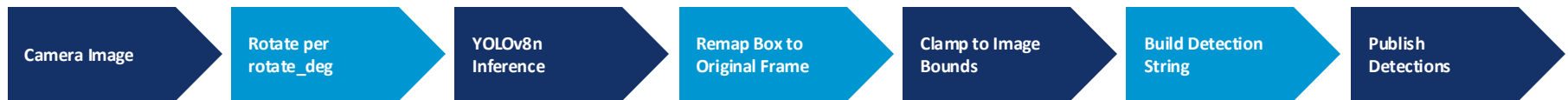
Training Environment

Google Colab (NVIDIA Tesla T4 GPU)

YOLO Model Training Dynamics



Privacy Object Detection Node (YOLO Model Inference)



- One detector instance per left/right torso mounted RGB camera.
- Rotates the frame matching the camera's physical mount.
- Uses the trained YOLO model to detect objects.
- Filters detections using a confidence threshold.
- Detected boxes are mapped back into the original (unrotated) image frame, since depth camera data is aligned to that frame.
- Boxes are clamped, so that the out-of-range coordinates never reach downstream nodes.
- Each detection is packed as a 9-field string containing class, confidence, bounding box coordinates, camera id etc.
- Detections are published to ROS for 3D position estimation using depth data.

LEFT Torso RGB Camera Topics

Input: `/torso_cam3d_left/rgb`
Output: `/privacy_detections_left`



RIGHT Torso RGB Camera Topics

Input: `/torso_cam3d_right/rgb`
Output: `/privacy_detections_right`

Depth Estimation & 3D Object Localization



- YOLO provides the 2D bounding box of the detected privacy-sensitive object.
- The center pixel (u, v) of the bounding box is used to look up depth.
- Depth comes from the RGB-D camera, not from YOLO.
- CameraInfo provides the intrinsic parameters f_x , f_y , c_x , c_y .

- A depth patch around (u, v) is sampled; the median valid depth is used.
- Back-projected to 3D: $X = (u - c_x)Z / f_x$, $Y = (v - c_y)Z / f_y$, $Z = \text{depth}$.
- The 3D point is transformed from the camera frame to /map via ROS TF.
- Tracks are maintained with position smoothing and range-jump checks.

LEFT camera

/torso_cam3d_left/depth_registered/image_raw
/torso_cam3d_left/rgb/camera_info

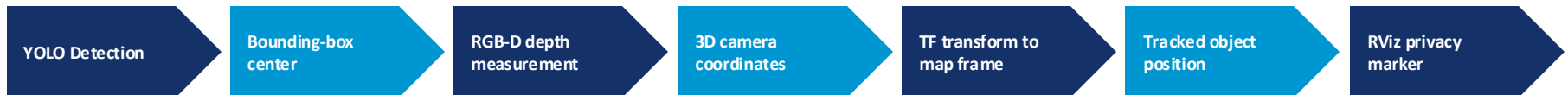


RIGHT camera

/torso_cam3d_right/depth_registered/image_raw
/torso_cam3d_right/rgb/camera_info

Metric depth + camera intrinsics → 3D position

Privacy Marker Generation



- `privacy_detector.py` publishes detections separately for the left and right cameras.
- `object_position.py` uses the corresponding RGB-D depth image and `CameraInfo` intrinsics.
- Depth is sampled around the bounding-box center using valid depth values.
- Object tracks receive persistent track IDs.
- `privacy_marker.py` creates a spherical RViz marker at the localized map position.

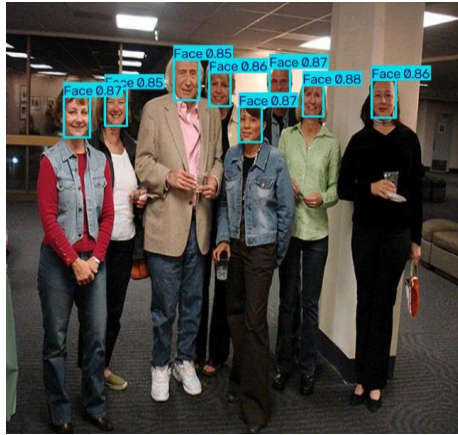
- The 3D point is back-projected using camera intrinsics and transformed directly from the camera optical frame into the “map” frame via TF.
- LEFT camera markers are RED; RIGHT camera markers are BLUE.
- A separate `TEXT_VIEW_FACING` marker displays the detected class, e.g. “Privacy Sensitive Object: Face”.
- Markers persist until the tracked object becomes stale; stale markers are explicitly deleted.



Markers visualize the privacy-sensitive location — the system records/displays it; it does not alter robot navigation or motion planning.

Results

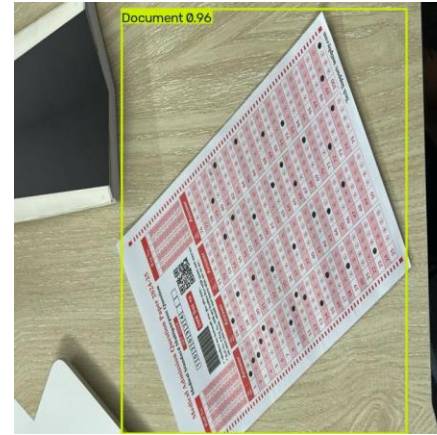
YOLO Model Results (Real Life Images)



Face



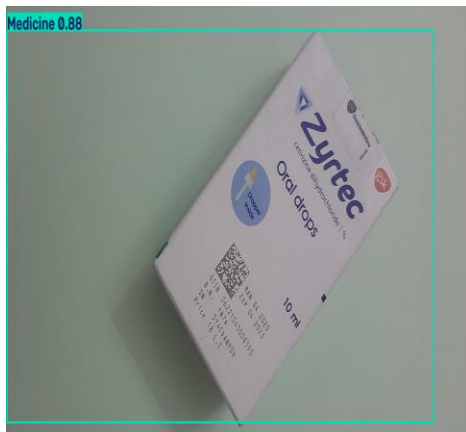
Display Monitor



Document



ID Card



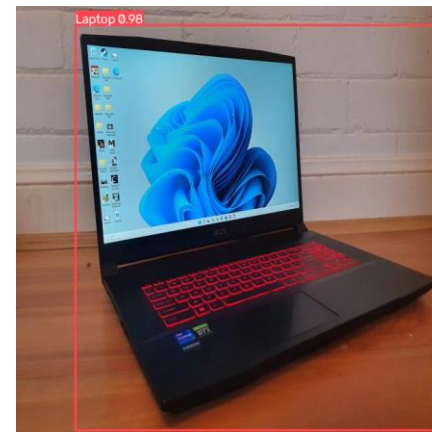
Medicine



Debit / Credit Card

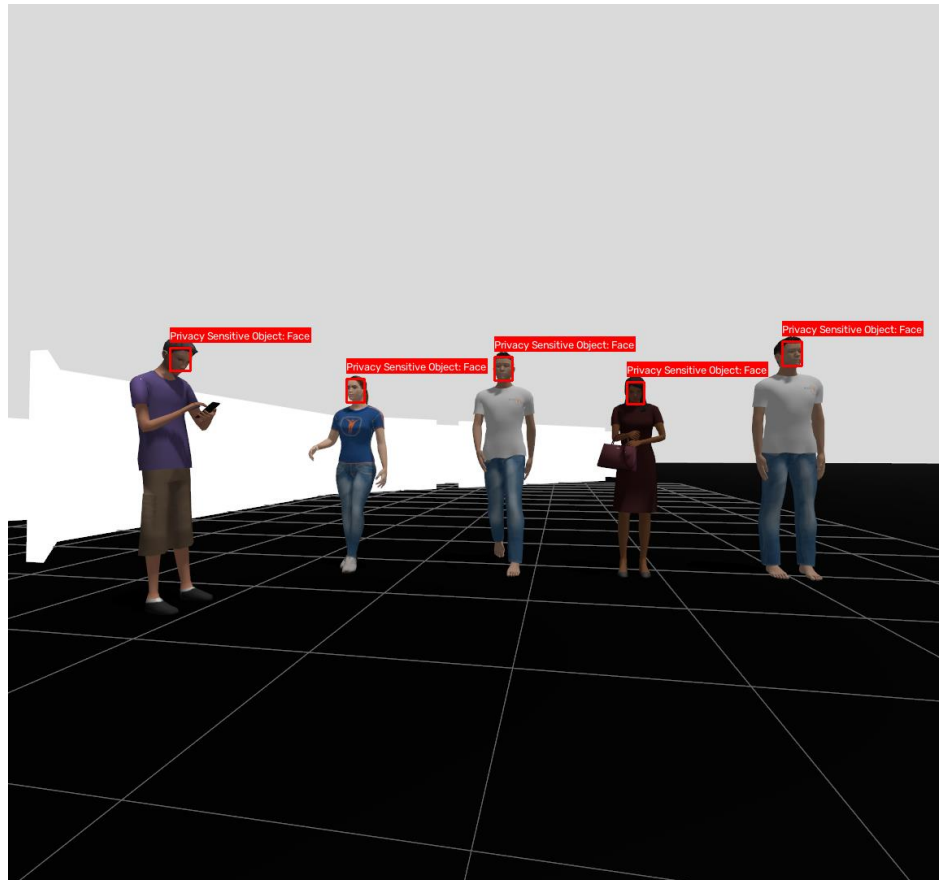


Car Number Plate

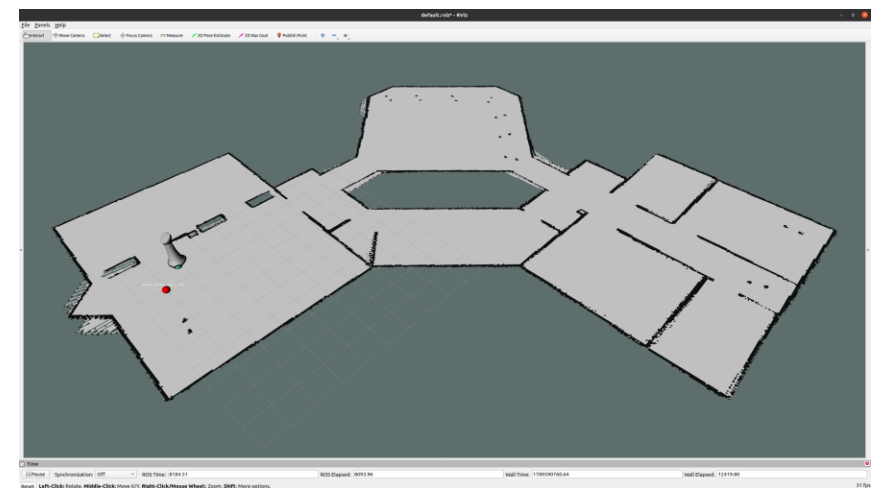
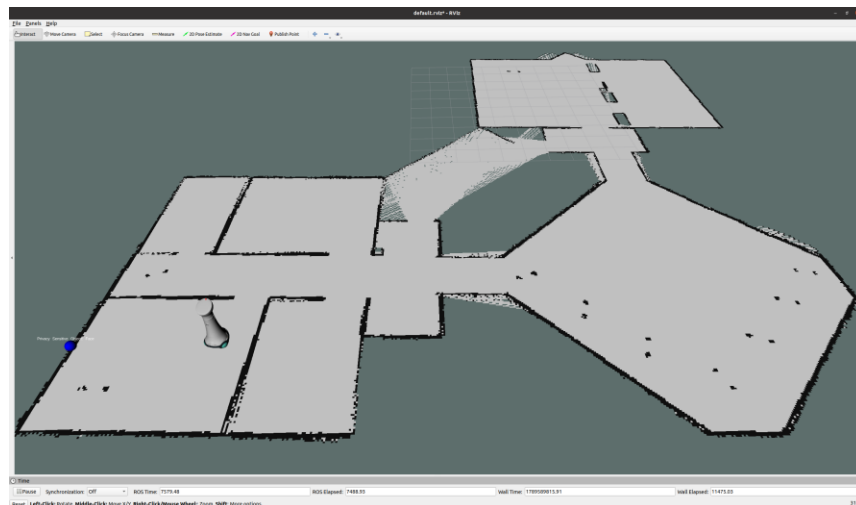
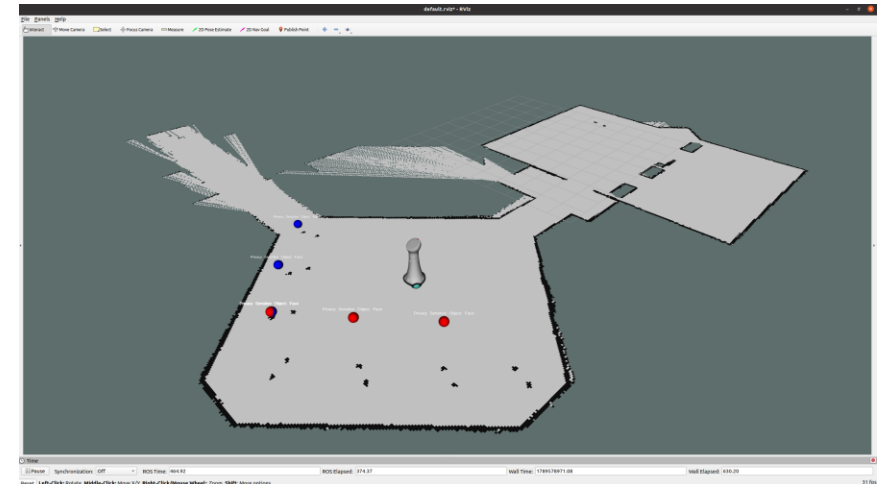
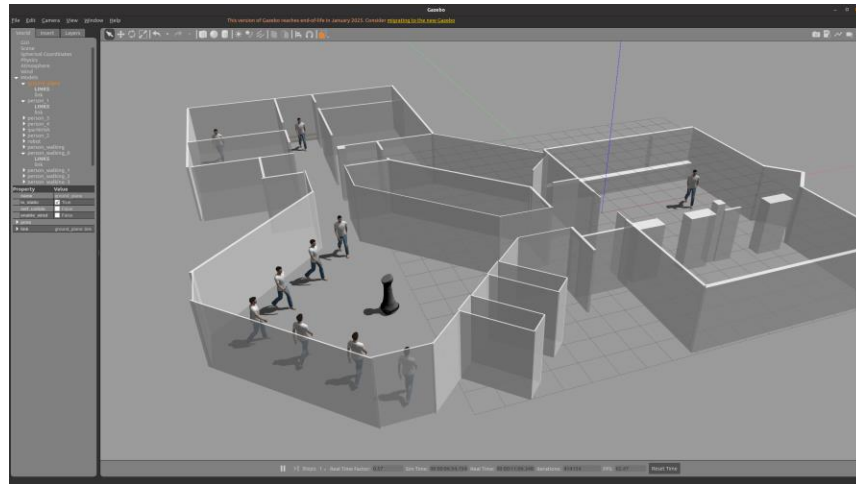


Laptop

YOLO Model Results (Simulator Environment)



Privacy-aware Environment Map



Limitations & Future Work

CURRENT LIMITATIONS

2D Mapping: GMapping represents the environment as a 2D occupancy grid; vertical information is lost.

SLAM & Exploration: Particle depletion can occur during sharp turns; mapping currently requires manual teleoperation.

Detection & Localization: Fixed YOLO classes and variable viewpoint, lighting, occlusion, and depth quality can affect results.

Simulation & Privacy Action: Evaluation is simulation-based and largely qualitative; privacy locations are visualized but do not yet change robot behavior.

FUTURE WORK

Better Mapping: Explore Cartographer and 3D mapping such as OctoMap.

Autonomous Exploration: Replace manual teleoperation with frontier-based exploration and ROS navigation.

Better Detection & Localization: Expand classes/data, improve filtering, multi-camera fusion, and depth/uncertainty handling.

Act on the Privacy Map: Use privacy-aware costmaps and potentially modify camera behavior around sensitive regions.

Conclusion

What We Delivered

✓ **Care-O-Bot Familiarization:** Learned the Care-O-Bot 4 platform and its Gazebo simulator.

✓ **LiDAR-Based Mapping:** Built a 2D environment map with GMapping SLAM from the unified laser scan.

✓ **Privacy-Sensitive Detection:** Trained YOLOv8 to recognize 8 privacy-sensitive object classes from RGB cameras.

✓ **Object Marking on Map:** Localized detections in 3D and rendered them as persistent privacy markers in RViz.

The Deeper Question to think about for future work:

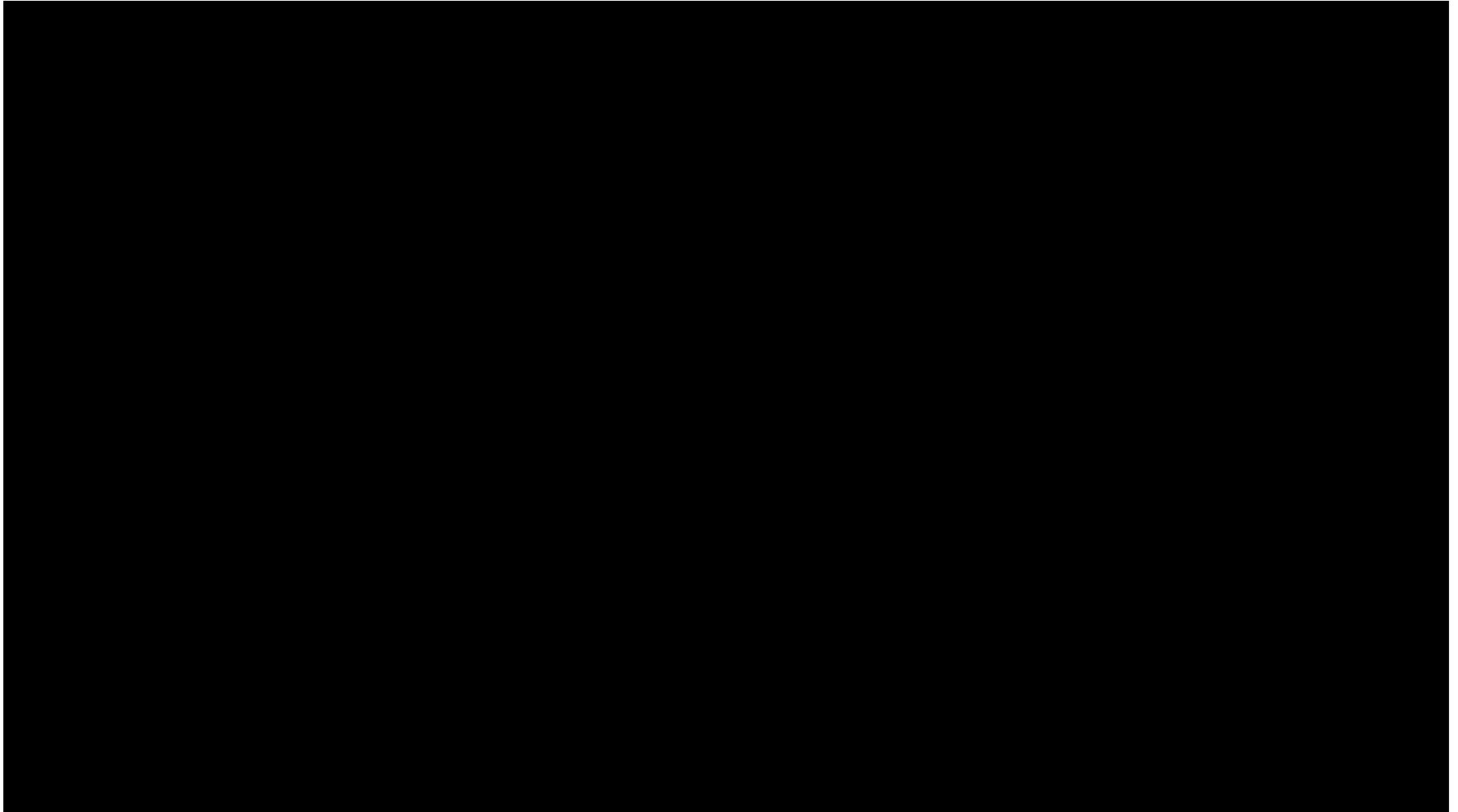
It's not "Can a robot detect and locate privacy-sensitive objects?"

It's: "How can a robot detect and locate privacy-sensitive objects without observing, capturing, or analyzing it (i.e. without compromising privacy itself)?"

References

- Fraunhofer IPA Care-O-bot, <https://www.care-o-bot.de/de/care-o-bot-4.html>
- Matic Robots, <https://maticrobots.com/>
- MathWorks, <https://www.mathworks.com/discovery/slam.html>
- S. Thrun, W. Burgard, D. Fox, Probabilistic Robotics, MIT Press, 2005.
- G. Grisetti, C. Stachniss, W. Burgard, "Improved Techniques for Grid Mapping With Rao-Blackwellized Particle Filters," IEEE Transactions on Robotics, 2007. GMapping / SLAM
- Open Source Robotics Foundation, gmapping: slam_gmapping, ROS Documentation. ROS GMapping implementation
- J. Redmon et al., "You Only Look Once: Unified, Real-Time Object Detection," IEEE CVPR, 2016. YOLO object detection
- I. Kostavelis, A. Gasteratos, "Semantic Mapping for Mobile Robotics Tasks: A Survey," Robotics and Autonomous Systems, 2015. Semantic mapping.
- M. Rueben et al., "Themes and Research Directions in Privacy-Sensitive Robotics," IEEE ARSO, 2018. Privacy-aware robotics
- T. Foote, "tf: The Transform Library," IEEE TePRA, 2013.
- YOLOv8, <https://yolov8.com/>
- YOLOv8 Ultralytics, <https://docs.ultralytics.com/models/yolov8>

Project Video



Questions?

